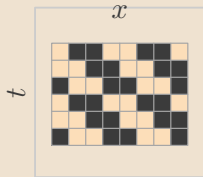


Selecting a relative-periodic domain with Bernoulli NML

1 Observed spacetime

Choose a candidate period and shift.
Here $p = 2$ and $s = 1$.



The observed CA spacetime U is the pattern to be explained.

2 Orbit classes

Translate by $(t, x) \mapsto (t + p, x + s)$.
Each orbit gives one template bit.



The orbit partition shrinks the spacetime block to a smaller template.

3 Majority-vote fit

Count ones and zeros on each orbit.
Majority vote sets the template bit.

orbit	count	vote
1	5 / 1	→
2	1 / 5	→
3	4 / 2	→
4	2 / 4	→

The voted bits form $B^*(p, s)$,
and $M = U \oplus B^*(p, s)$
is the residual mask.

4 Score and select

Score each candidate with Bernoulli NML.
Smaller totals are better.
 $\text{NML}(p, s) = \text{fit} + \text{penalty}$

Fit pure orbits reduce fit

Penalty extra bits raise cost

Choose the candidate with the smallest total NML.

Output: the selected scaffold and its residual mask.